



Accenture and Technology News Insights Engine AI Studio Final Presentation

UCLA
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Introductions



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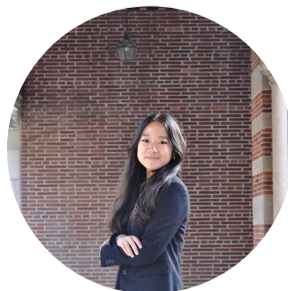
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Presentation Agenda

1. Data Understanding & Preparation
2. Technology News Sentiment Analysis & Forecasting
3. Technology News Preprocessing Pipeline & Topic Modeling
4. Technology News Entity Relationship Mapping
5. Technology News Recommendation System



AI Studio Project Overview



The Technology News Insights Engine analyzes various technology news articles to uncover trends in tech products and services for advising tech startups on how to grow their businesses.





Our Goal

1. Organize technology news articles into each technology category
2. Assign a topic name to each group of technology news articles
3. Uncover short terms trends in the tech industry and development
4. Identify emerging trends, key players, and potential partnerships within the tech industry
5. Enable data-driven insights from technology news articles to inform strategic decisions based on a recommendation system





Business Impact

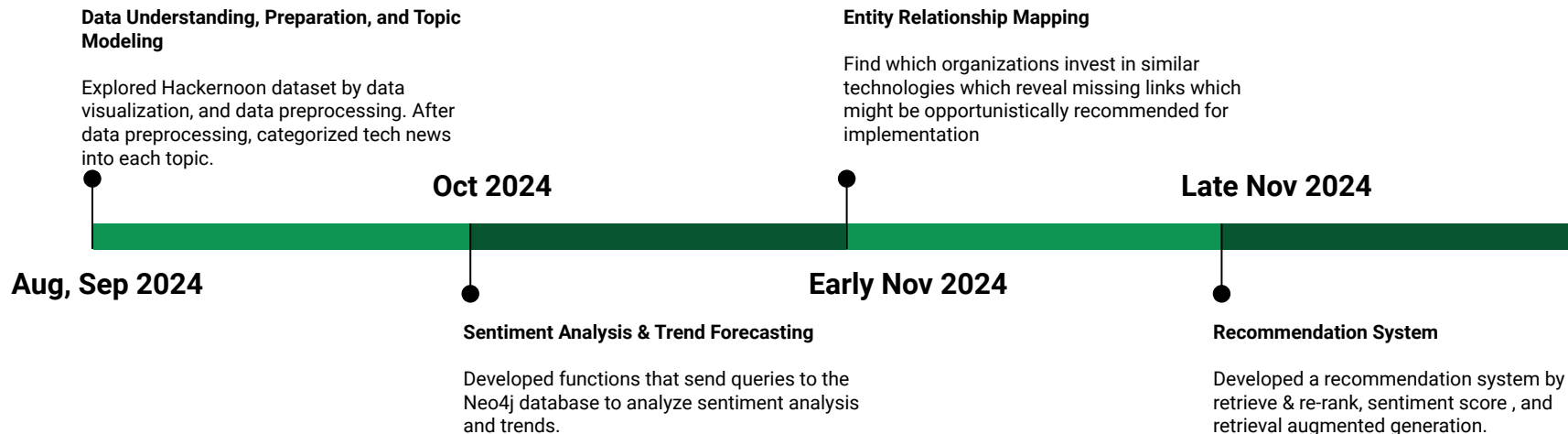
Aid technology consultants at Accenture:

- Assess the tech resources available in their engine or database to gain insights into existing capabilities and infrastructure
- Analyze emerging trends and support tech startups in leveraging new market opportunities for growth and innovation
- Efficiently gather the latest information on specific technologies tailored to the types of technologies clients work with
- Explore strategic partnerships and identify growth opportunities by evaluating potential collaborations, partnerships, or acquisitions with organizations that align with business objectives





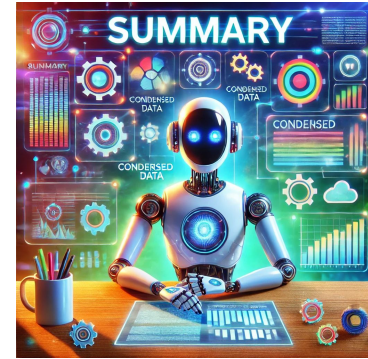
Our Approach



Summary of Insights & Key Findings



- Organize 80,000 Technology News articles into **11** distinct technology categories
- Analyzed sentiments of top companies to advise tech startups and Accenture where to allocate their resources
- Visualized trends of key words and identified top 5 companies (Google, Microsoft, Sony, Epic Games, Animoca Brands) by article volume
- Depicted which companies or organizations invested in similar technologies, focusing on the industries
- One of the response from the recommendation model:
 - **Cloud accounting software platforms** help small businesses manage finances efficiently through features like automation, integration, real-time reporting, and sustainability benefits, while addressing challenges like security, customization, and internet reliance, with major providers such as QuickBooks and FreshBooks offering scalable solutions that promote environmental and economic sustainability.





Data Understanding & Data Preparation



Resources We Leveraged

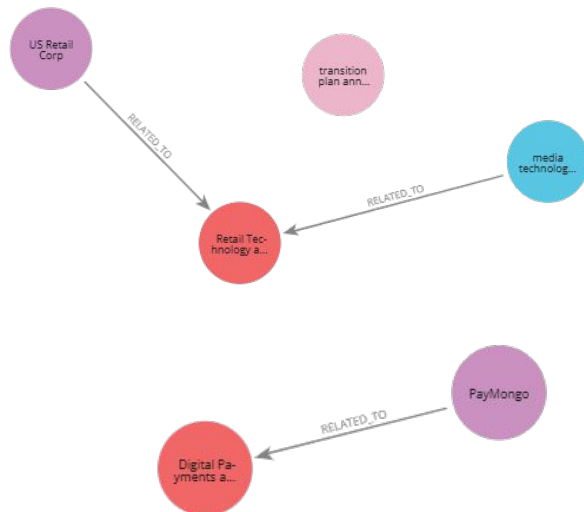
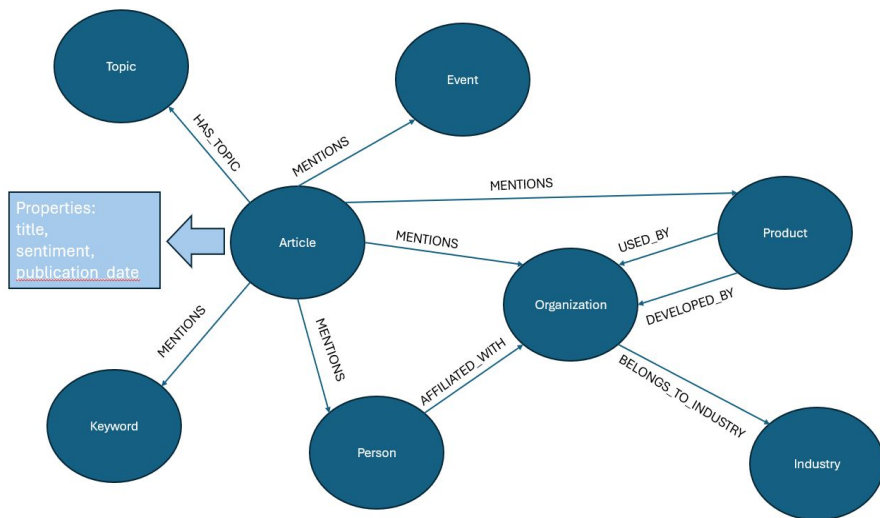
- Coding IDE
 - Google Colab
- Project Management tool
 - Trello
- Database
 - Neo4j Graph Database
- Python Libraries
 - Pandas, Numpy, Matplotlib, Seaborn, Scikit-Learn
 - NLTK, Gensim, Groq, Scipy, Network, BERTopic, HDBSCAN, UMAP, KeyBERTInspired, MaximalMarginalRelevance, PartOfSpeech
 - Transformers, Sentence Transformers, Llama-index, langchain, TextBlob



Data Overview



- Our team was provided a **Neo4j knowledge graph** that was built from tech news articles
- **Initial approach:** Scraping and summarizing web article data
- **Challenges:** Google Colab's computing power made this approach unviable
- **New approach:** Utilizing a knowledge graph and focusing more on data analysis





Data Analysis

- Created Python functions to send queries to the Neo4j database to access data

```
# Function to retrieve sentiment trends for multiple companies between 2022 and 2024
```

```
def get_sentiment_trend_companies(companies):
```

```
    with driver.session() as session:
```

```
        result = session.run(
```

```
            """
```

```
            MATCH (o:Organization)-[:MENTIONS]-(a:Article)
```

```
            WHERE o.name IN $companies
```

```
            AND a.publication_date IS NOT NULL
```

```
            AND a.sentiment IS NOT NULL
```

```
            AND a.publication_date >= date('2022-01-01')
```

```
            AND a.publication_date <= date('2024-12-31')
```

```
            WITH o.name AS company,
```

```
                 date.truncate('month', a.publication_date) AS month_year,
```

```
                 a.sentiment AS sentiment
```

```
            RETURN company,
```

```
                 month_year.year AS year,
```

```
                 month_year.month AS month,
```

```
                 AVG(sentiment) AS avg_sentiment
```

```
            ORDER BY year, month, company;
```

```
            """, {"companies": companies}
```

```
        )
```

```
# Convert results to DataFrame
```

```
data = [{"company": record["company"], "year": record["year"], "month": record["month"], "avg_sentiment": record["avg_sentiment"]} for record in result]
```

```
df = pd.DataFrame(data)
```

```
return df
```

- Finds organizations in articles
- Filters by date and sentiment
- Aggregates sentiment by month



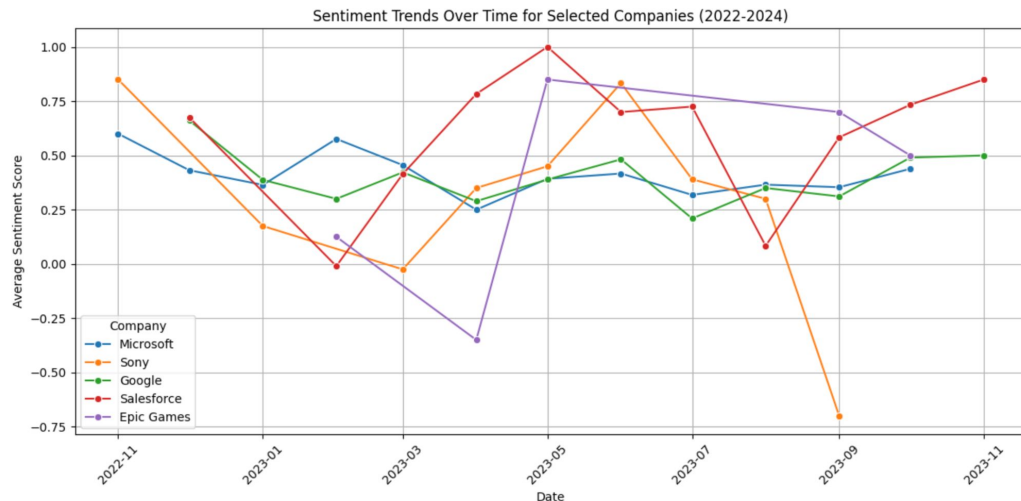
Technology News Sentiment Analysis & Trend Forecasting



Sentiment Analysis

- **Approach:** Created a function to plot the average sentiment of a company each month
 - Sent a query to the Neo4j database to grab the average sentiment of the companies every month and plotted the data using a line graph
 - Graphed sentiment trends for top companies between 2022 - 2024
- **Further Development:** Enables Accenture consultants to use our function by typing in any organization name, which our function will plot directly

```
def get_sentiment_trend_companies(companies):  
    with driver.session() as session:  
        result = session.run(  
            """  
            MATCH (o:Organization)-[:MENTIONS]-(a:Article)  
            WHERE o.name IN $companies  
              AND a.publication_date IS NOT NULL  
              AND a.sentiment IS NOT NULL  
              AND a.publication_date >= date('2022-01-01')  
              AND a.publication_date <= date('2024-12-31')  
            WITH o.name AS company,  
                  date.truncate('month', a.publication_date) AS month_year,  
                  a.sentiment AS sentiment  
            RETURN company,  
                    month_year.year AS year,  
                    month_year.month AS month,  
                    AVG(sentiment) AS avg_sentiment  
            ORDER BY year, month, company;  
            """, {"companies": companies}  
        )
```

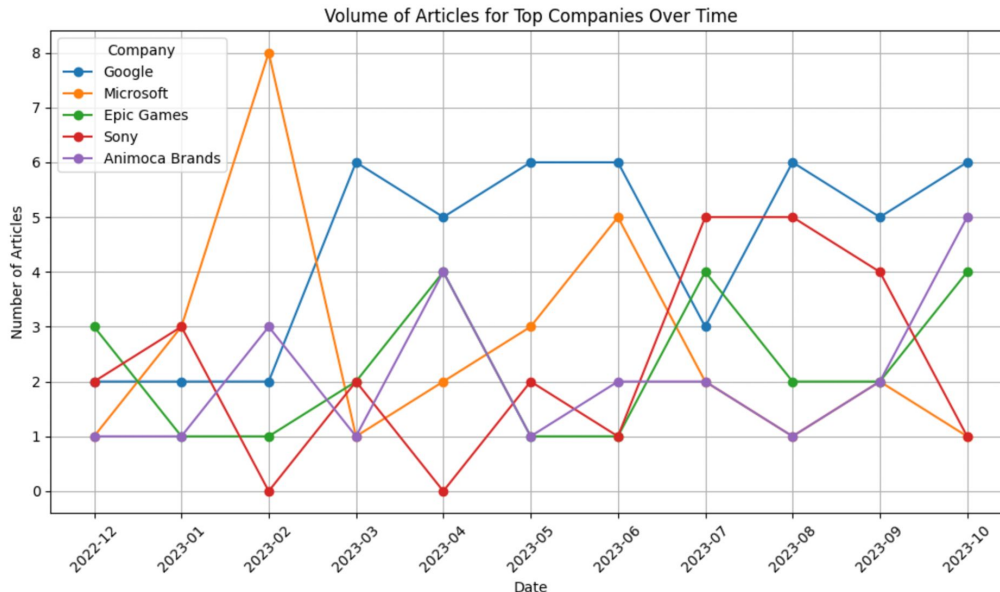




Trend Forecasting

- **Approach:**

- Identified top 5 companies with the highest article volume using `value_counts()`
- Visualized the article volume trends over time for top companies between 2022-2024

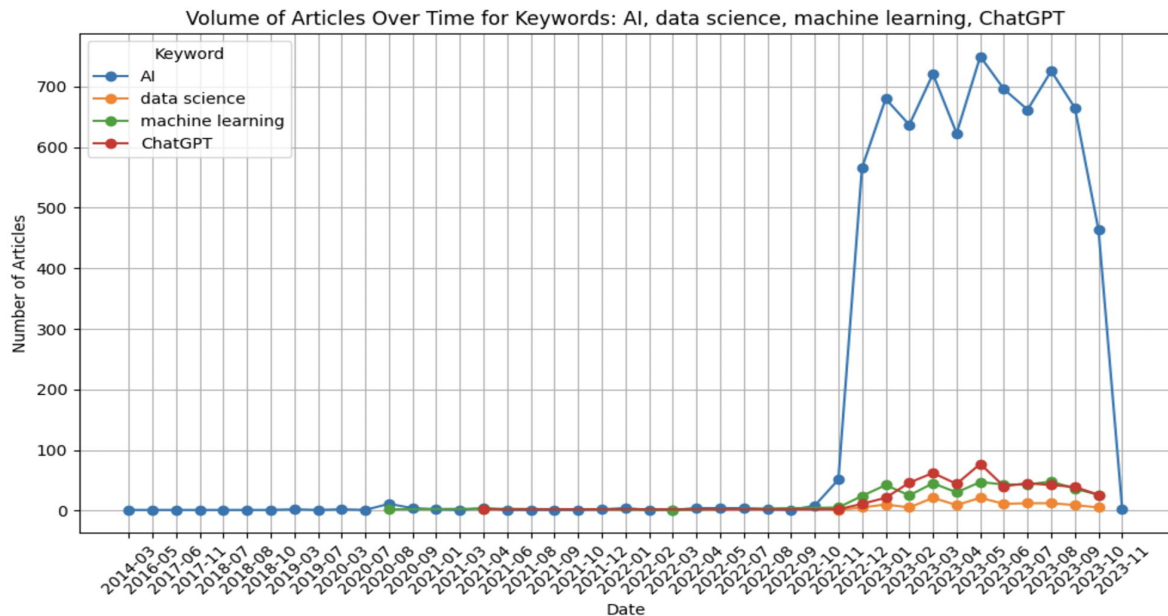


Trend Forecasting



- **Approach:**

- Defined a list of target keywords to search for in articles
- Filtered articles containing each keyword and aggregated the count of articles mentioning the keywords on a monthly basis
- Plotted time-series data for each keyword





Recommendations

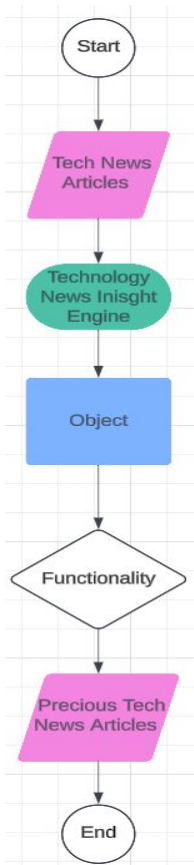
From the sentiment analysis and trends, we see from 2022-2024:

- Recommend tech startups to **move towards AI products and services** as there was an large mention of AI in articles in 2023
 - The trend is expected to be consistent in 2024
- Look into **Google and Microsoft's products** for building a base for startups
 - Ex: Google Cloud Platform/Workspace, Microsoft Azure
 - Salesforce potentially due to upward trend at the end of 2023
 - Sentiment analysis scores are understandable due to the company's reputation
- **Company Trend Forecasting**
 - Allow Accenture to identify potential partnerships or collaborations with leading companies in the media
 - Companies receiving significant media attention may be expanding or innovating, allowing Accenture to pitch services that align with their priorities



Technology News Insight Engine Structure Overview

Basic Engine Structure



Raw Data: Technology News Articles

Engine: Engine has the ability of data annotation and recommendation based on a user query

Object / Class: groups instructions / codes to complete a specific task

Functionality: the abilities of an object

Precious technology news articles or other output



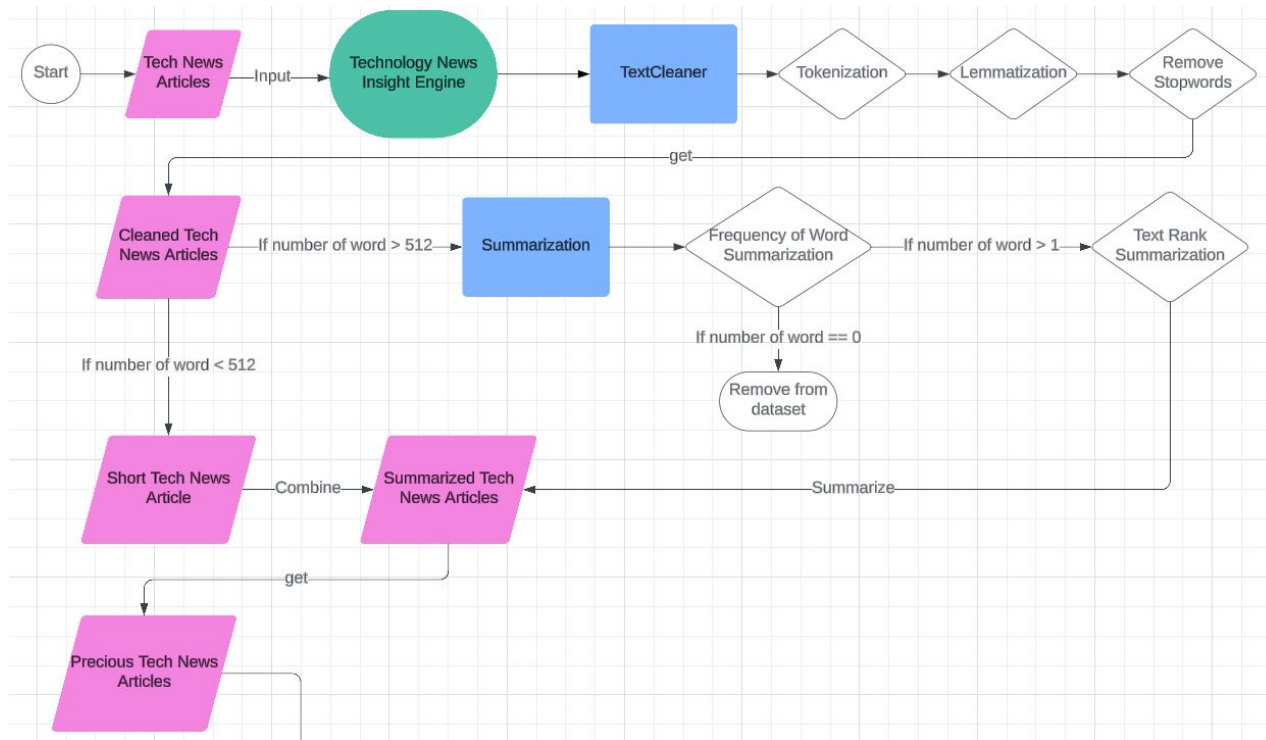
Technology News Preprocessing & Topic Modeling Pipeline

Preprocessing Pipeline



TextCleaner: Split sentences, convert words to their root forms, and remove meaningless words.

Summarization: Summarize by frequency of word and TextRank Summarization





Summarization Methods

Frequency of word summarization:

1. Calculate the frequency of each word in an article
2. Assign a score to each sentence based on the frequency of each word in each sentence
3. Select the sentences has a score greater than the median score over all sentences

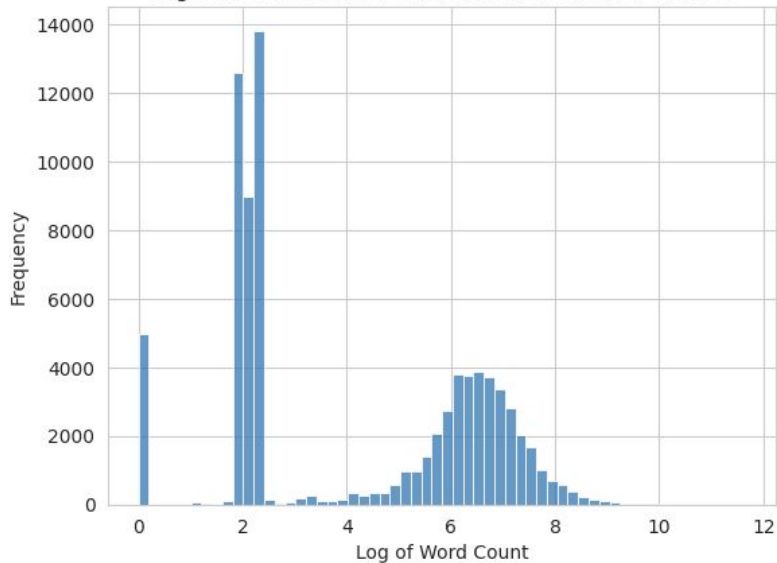
TextRank Summarization:

1. Calculate the similarity scores between each sentence
2. Pick the sentences has a similarity score greater than the median similarity score over all sentences

Preprocessing Results

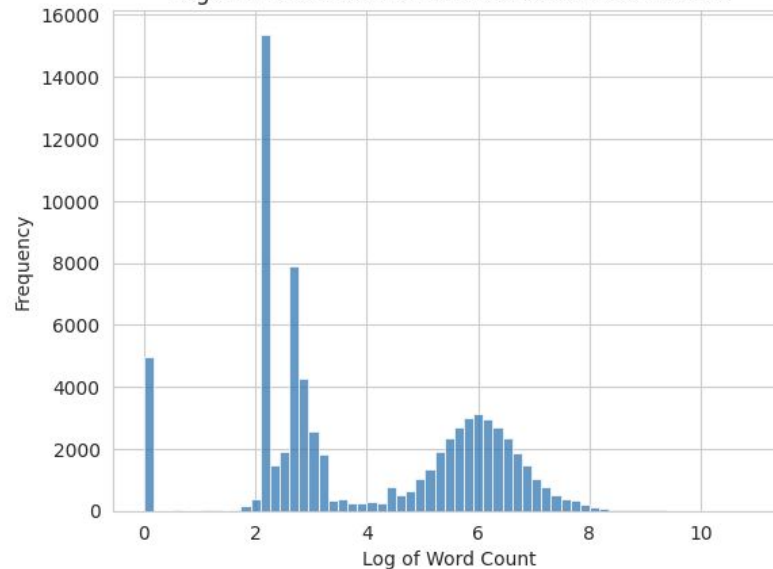


Log of Word Count Distribution in uncleaned articles



min	max	median	std
0	115,782	10	1338

Log of Word Count Distribution in cleaned articles



min	max	median	std
0	10,785	23	168

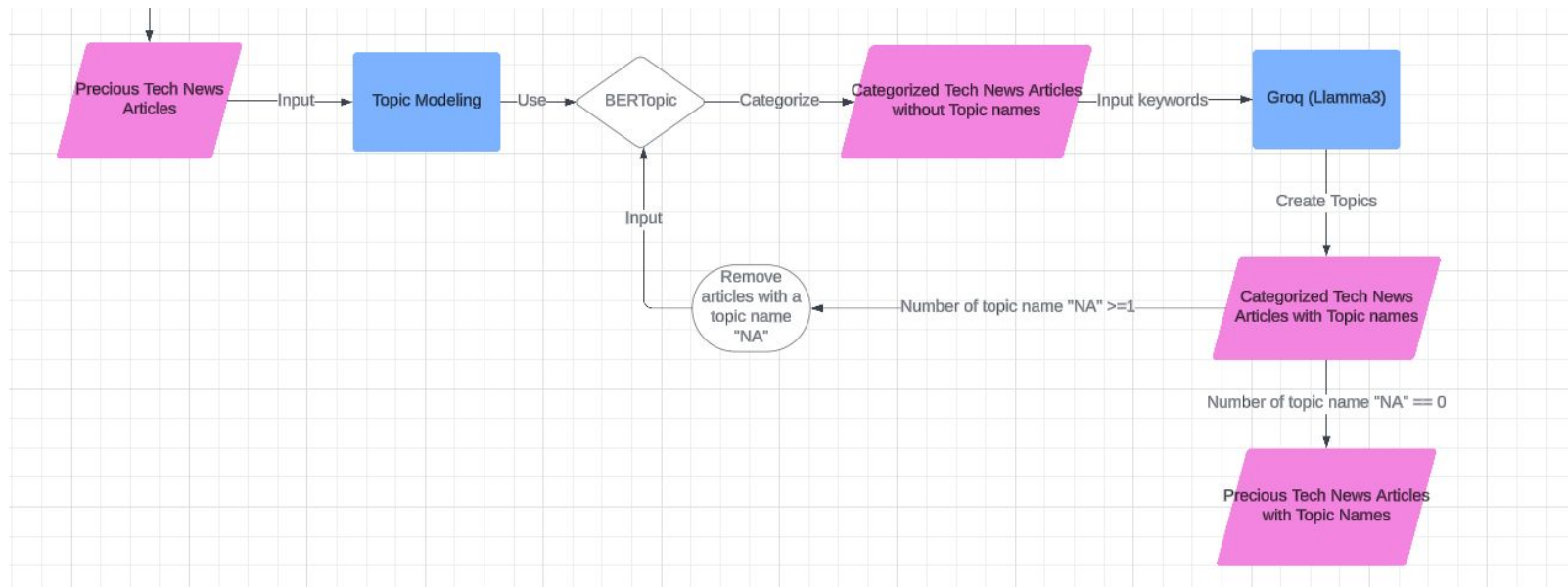


Topic Modeling Pipeline

Goal: Remove articles related to consumer technology and error messages

BERTopic: Python library which can help us to categorize and extract keywords from tech news articles

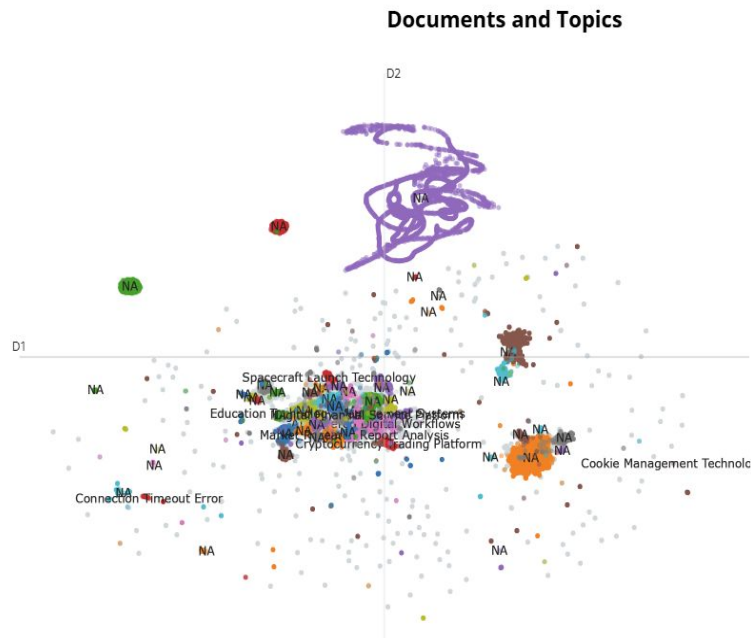
Groq: free AI Cloud where we can use various fine-tuned large language model (e.g. Llama3, gemma, whisper)



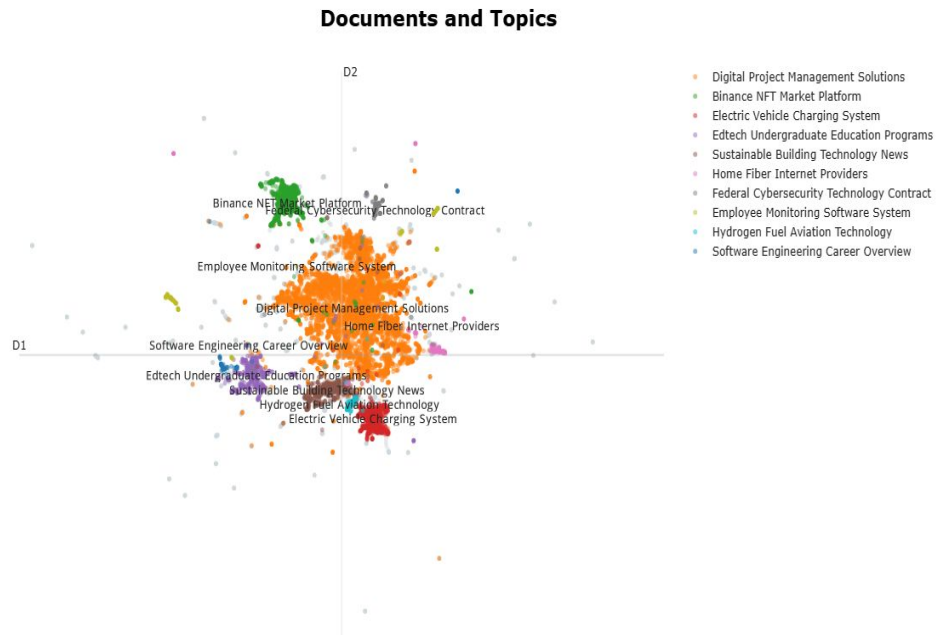


Topic Modeling Results

80,000 Tech News Articles before filtering



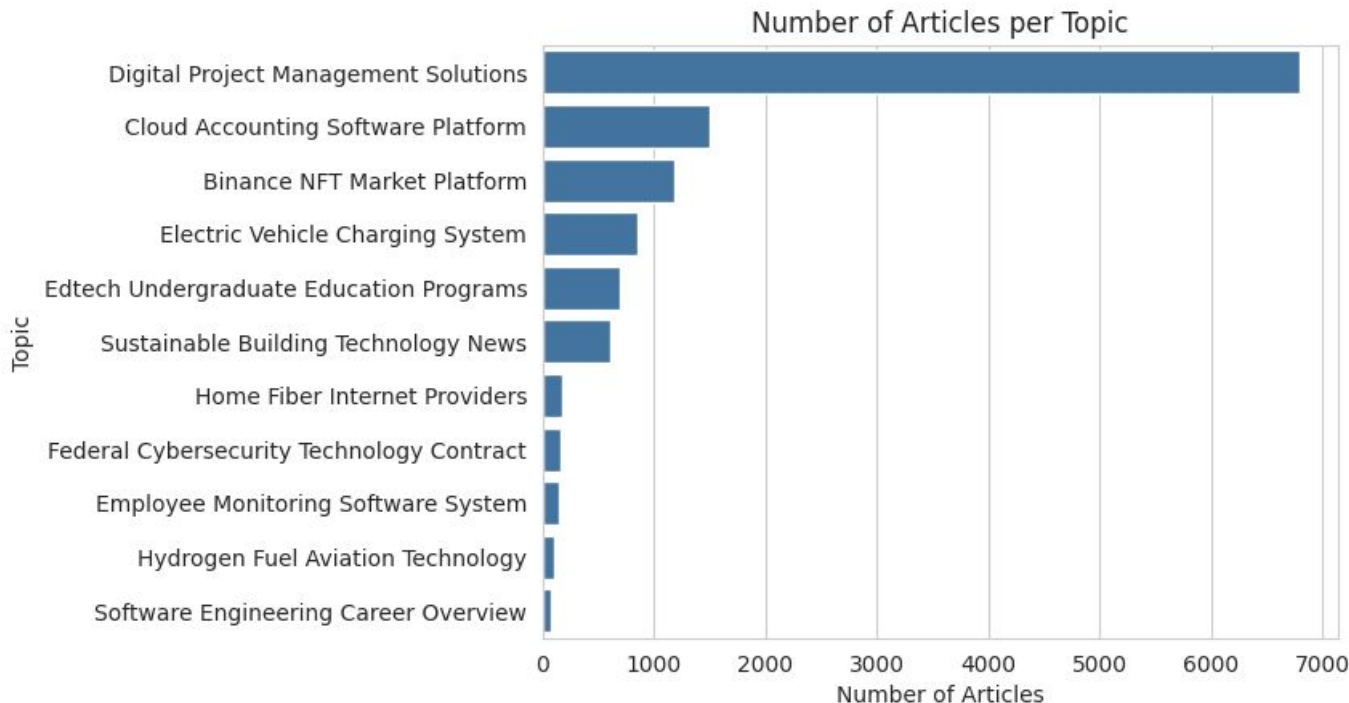
12,269 Tech News Articles after filtering





Topic Modeling Results

The topic “Digital Project Management Solutions” has the greatest number of articles in the engine





Entity Relationship Mapping



Topic Entity Relationship Mapping

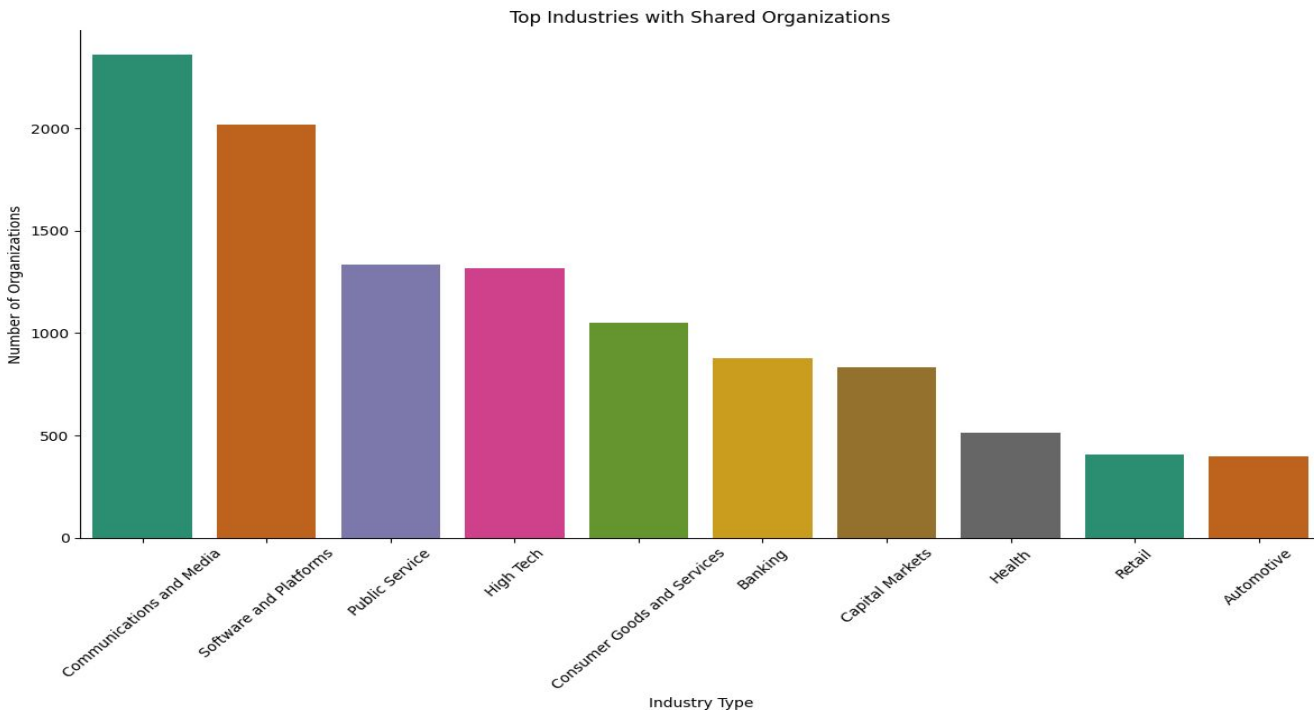
- Categorized organizations based on the industries or technologies they invest in with the first column listing industry types and the second column showing the corresponding organizations

	industry_type	orgs
0	Aerospace and Defense	[Organization1, U.S. Department of Defense, NA...
1	Automotive	[Organization2, Organization3, BYD Co., Merced...
2	Banking	[Organization3, Goldman Sachs, Inter-American ...
3	Capital Markets	[Organization4, Goldman Sachs, SoftBank, ICE D...
4	Chemicals	[Organization5, 3M, Infor, Honeywell, Snapdrag...
5	Communications and Media	[Amazon Web Services, IDC, Network World, New ...
6	Consumer Goods and Services	[Alibaba, Realt, Roofstock, Organization5, New...
7	Energy	[KCA Deutag, LHR Marine, International Renewab...
8	Health	[Organization2, Organization4, Tempus, Therano...
9	High Tech	[Google Cloud, Amazon Web Services, Alibaba, I...
10	Industrial	[Dandara, HVAC, Otis, Clearwater Electrical, C...
11	Insurance	[Berkshire Hathaway, Experian, IBM, Aura, Iden...
12	Life Sciences	[JB Pharma, Cardea Bio, MimiVax, Workday, MIT...
13	Natural Resources	[FLSmidth & Co. A/S, ArcelorMittal Holdings AG...
14	Public Service	[New Orleans Pelicans, Boston Celtics, U.S Gov...
15	Retail	[Alibaba, EG Group, Asda, Organization5, Shopl...
16	Software and Platforms	[Google Cloud, Amazon Web Services, Microsoft ...
17	Travel	[Sarasota Bradenton International Airport, SRQ...
18	US Federal Government	[NASA, FCC, US Treasury, Department of Treasur...
19	Utilities	[Richard Irvin, Bahamas Power & Light (BPL), W...



Topic Entity Relationship Mapping

- Developed a bar graph to depict which companies and organizations invested in similar technologies, focusing specifically on the top industries.





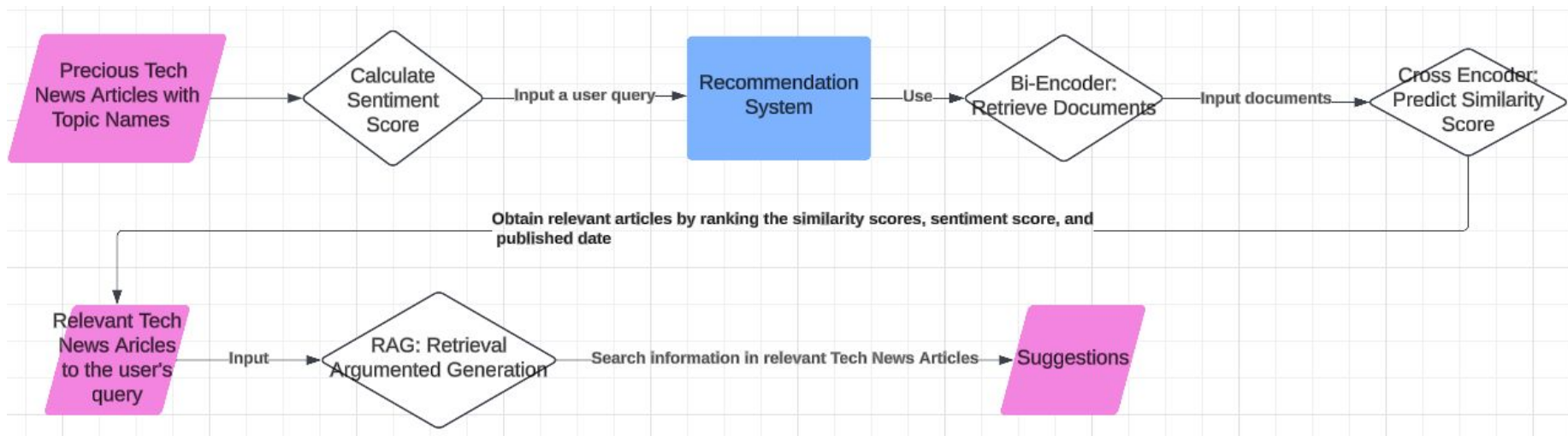
Technology News Recommendation System



Recommendation System Overview

Goal: Input a list of relevant articles to a user's query in a chatbot model and receive a suggestion from the model

Recommendation system: consists of Bi-Encoder and Cross Encoder from sentence transformers and Retrieval Argumented Generation (RAG)





Recommendation System Example (1)

Question: What is the initiatives of cloud accounting software platform?

Answer: Cloud accounting software platforms streamline financial management with features like

1. Automated payroll
2. Time-saving tools
3. Seamless system integration
4. Customizable dashboards
5. Invoicing, tax management
6. Flexible subscription plans, and free trials.



Recommendation System Example (2)

Question: Who are the biggest providers of cloud accounting software platform and what type is being used?

Answer:

1. **QuickBooks:** A widely used platform for small businesses, offering invoicing, expense tracking, inventory management, and payroll processing.
2. **Xero:** Known for invoicing, expense tracking, and project management, with integration to payment gateways and banking apps.
3. **FreshBooks:** Tailored for freelancers and small businesses, offering invoicing, time tracking, expense tracking, and integration with payment gateways and project management tools.
4. **Zoho Books:** Provides invoicing, expense tracking, and project management, and integrates with Zoho's CRM and inventory management tools.
5. **Wave:** A free platform for small businesses and freelancers, offering invoicing, expense tracking, payment processing, and integration with payment gateways and banking apps.



More Questions

Questions:

1. What are the strength, weaknesses, opportunity, and threats for cloud accounting software platform and each company?
2. What are the primary challenges or risks faced by clients when adopting or using a cloud accounting software platform?
3. How much money is being invested in a cloud accounting software platform? How long are they anticipated to take?
4. What is the impact on sustainability metrics in a cloud accounting software platform? How will this impact jobs?





Final Thoughts



Potential Next Steps

- Integrate all our codes to create a complete technology news insight engine that have **all the functionalities**
 - Sentiment Analysis and Forecasting
 - Topic Modeling
 - Entity Relationship Mapping
 - Strategic Recommendation System
- Add web scraping functionality
- Replace the summarization methods with **SmolLM-v2** on Transformer
- Fine-tuned **Llama2-7b-chat-hf** model used by recommendation system and **SmolLM-v2**



Thank you!



Questions?



What We Learned

- Topic Model



Appendix



Recommendation System Example (3)

Question: What are the strength, weaknesses, opportunity, and threats for cloud accounting software platform and each company?

Answer:

1. **Strengths:**

- Cloud accounting software is user-friendly, offers real-time reporting, is scalable, facilitates collaboration, and ensures automatic data backups.

2. **Weaknesses:**

- It may raise security concerns, requires stable internet connectivity, lacks full customization, and can be costly for some businesses.

3. **Opportunities:**

- There is growing demand, integration potential with other business tools, mobile access for flexibility, and opportunities to expand into new markets.

4. **Threats:**

- The market is highly competitive, and regulatory changes may pose challenges or necessitate software updates.



Recommendation System Example (4)

Question: What are the primary challenges or risks faced by clients when adopting or using a cloud accounting software platform?

Answer:

- Clients may face **challenges** when adopting cloud accounting software, including:
 1. Concerns over data security and privacy
 2. Loss of control over financial data, and dependence on internet connectivity
 3. Compliance with regulations
 4. Difficulty integrating with other systems
 5. Concerns about cost vs. value
 6. Limited customization options, and the need for support and training.



Recommendation System Example (5)

Question: How much money is being invested in a cloud accounting software platform? How long are they anticipated to take?

Answer:

- Cloud accounting software platform will take around **12 months** to develop, with an estimated investment of **\$500,000**. However, it is important to note that these estimates are based on the information provided and may not reflect the actual time and investment required for the development of the platform.



Recommendation System Example (6)

Question: What is the impact on sustainability metrics in a cloud accounting software platform? How will this impact jobs?

Answer:

1. **Reduced Carbon Footprint:**
 - a. It lowers carbon emissions by eliminating the need for physical hardware like servers and data centers.
2. **Energy Efficiency:**
 - a. The use of energy-efficient technologies minimizes energy consumption and emissions.
3. **Sustainable Data Centers:**
 - a. Cloud providers are investing in eco-friendly data centers powered by renewable energy sources.
4. **Remote Work:**
 - a. Enables remote work, reducing commuting and cutting carbon emissions.
5. **Job Creation:**
 - a. Promotes economic growth by creating new jobs in software development, data analysis, and customer support.



Frequency of Word Summarization

1. Word Counting:
 - It tokenizes the article into individual words.
 - It creates a frequency table (freqTable) storing the count of each word (excluding common stop words).
2. Sentence Scoring:
 - It tokenizes the article into sentences.
 - For each sentence, it calculates a score based on the frequencies of the words present in that sentence using the freqTable.
 - Sentences containing more frequent words receive higher scores, indicating their potential importance.
3. Summarization:
 - It calculates the median sentence score.
 - It iterates through the sentences and selects those with scores significantly above the median (by a factor of 1.20 in this case).
 - These selected sentences are combined to form the summary.

TextRank Summarization



1. Sentence Embeddings:
 - The text is first split into individual sentences.
 - Each sentence is then converted into a **numerical vector** representation called an embedding. These embeddings capture the semantic meaning of the sentences.
2. Similarity Matrix:
 - A **similarity matrix** is constructed to represent the relationships between sentences.
 - Each element in the matrix indicates the similarity between two sentences based on their embeddings. Higher values denote stronger semantic similarity.
 - The `similarity_matrix` method in the code calculates this matrix using cosine similarity.
3. Graph Representation:
 - The similarity matrix is then interpreted as a graph where sentences are nodes and the similarity scores are edge weights.
 - This graph representation captures the interconnectedness of sentences based on their **semantic relationships**.
4. PageRank Algorithm:
 - The PageRank algorithm, originally used for ranking web pages, is applied to this sentence graph.
 - It iteratively **assigns scores** to each sentence (node) based on the scores of its connected sentences (neighbors).
 - Sentences with connections to high-scoring sentences receive higher scores, indicating their importance in the overall context.
5. Sentence Selection:
 - Finally, the sentences with the **high PageRank scores** are selected to form the summary.
 - These sentences are considered the most central and informative in the text, representing the main ideas.



Retrieval Augmented Generation

Retrieval-Augmented Generation (RAG) combines two key elements—retrieval and generation—to create smarter AI systems.

1. Retrieval: Imagine having a massive library of information. When you ask a question, instead of trying to remember everything, the system quickly looks up the most relevant articles to find useful information.
2. Generation: Once the system has the relevant information, it uses AI (like GPT) to write a clear response, combining the facts it found with its own language skills.